Further OEIS proofs and formalizations

Expanded proof package, version 2

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AI disclosure and review status. ChatGPT (OpenAI) generated the new proofs, exposition, Lean formalization, verification programs, and AI-assisted reviews in response to Dev Goyal's requests. Dev Goyal is the sole human author and initiated the work. **Independent human review is pending.** Upstream code and earlier mathematical results are credited separately. This document does not claim OEIS acceptance, peer review, or exclusive priority.

Abstract

We give formally checked proofs of seven statements labeled as conjectures in four retrieved OEIS entries: two parity assertions for A396093, two congruences each for A389474 and A389475, and the coefficient congruence for A396102. Two of the cubic assertions are direct instances of the standard Frobenius identity. The sequence-specific arguments give a complete period-ten parity classification, an identity for an entire square modulo nine, and a general family of inverse-iterate congruences. We construct the implicit integer series and prove their source equations and uniqueness. We also formalize a totient-period theorem for all factorial-weighted Stirling transforms and seven named applications. The latter are already covered by Danesh's 2026 theorem, with a stronger Carmichael bound, and are not counted as newly solved open problems. The archive retains all previous proofs, the previous manuscript, and the previous ZIP verbatim.

1 Scope, notation, and historical status

All series in this paper are formal. A normalized series means $U = x + O(x^2)$. We distinguish ordinary powers U^k from compositional iterates $U^{\circ k}$, with $U^{\circ 0} = x$. A congruence of series is coefficientwise, and $\mathbb{N}$ includes zero. In particular, no convergence assumption is required for any statement below.

The source pages were consulted on 24 October 2026. The four entries in Table 1 explicitly labeled the indicated statements as conjectures [1, 2, 3, 4]. That label is evidence about the retrieved entry, not a guarantee that no proof exists elsewhere. Our contribution is a reproducible proof package with exact mathematical scope. We make no global novelty claim. The two Frobenius consequences are standard algebra applied to the stated sequences.

The seven additional periodicity applications in Section 5 have a different historical status. Danesh's Theorem 4.1 and Corollary 4.2 already imply them [5]. They are included as formalizations and recovered corollaries of known mathematics. The package does not inflate its open-problem count by treating those seven cases, their general theorem, or its intermediate lemmas as new discoveries.

Entry	Statements addressed	Count
A396093	Even coefficients at even indices; exact condition for evenness at odd indices	2
A389474	Square coefficients above degree three vanish modulo nine; cube-power congruence modulo three	2
A389475	Cube substitution and the associated square identity modulo three	2
A396102	Every positive-index coefficient is one modulo three	1

Table 1: Seven newly added proofs of statements marked conjectural in the retrieved entries.

2 A396093: a complete parity classification

The original definition is

$$B(x) = \frac{x}{(1-x)^2}, \qquad A = B \circ B \circ B, \qquad a_n = [x^n]A \in \mathbb{Z}.$$

The offset is zero, so $a_0 = 0$. These definitions, rather than a conjectured coefficient pattern, are the starting point of the Lean construction.

Theorem 2.1. *For every $n \geq 0$,*

$$a_n \equiv \begin{cases} 1 \pmod 2, & n \equiv 1, 3, 7, 9 \pmod{10}, \\ 0 \pmod 2, & \textit{otherwise.} \end{cases}$$

Consequently a_{2n} is even for every $n \geq 0$. For $n \geq 1$,

$$a_{2n-1} \textit{ is even} \iff n \equiv 3 \pmod 5 \iff n = 5k - 2 \textit{ for some } k \geq 1.$$

Proof. Reduce coefficients to $\mathbb{F}_2$. In characteristic two, $B(x) = x/(1+x^2)$. If $T = P/D$ has zero constant coefficient and $D(0) = 1$, then

$$B(T) = \frac{PD}{D^2 + P^2}.$$

Every denominator has constant coefficient one and is therefore an invertible formal series. Applying this identity gives

j	numerator of $B^{\circ j}$	denominator
1	x	$1 + x^2$
2	$x + x^3$	$1 + x^2 + x^4$
3	$x + x^7$	$1 + x^2 + x^4 + x^6 + x^8$.

Multiplication of the last numerator and denominator by $1 + x^2$ gives the exact identity

$$A(x)(1 + x^{10}) = x + x^3 + x^7 + x^9 \qquad \text{in } \mathbb{F}_2[[x]]. \tag{1}$$

Since $(1 + x^{10})^{-1} = \sum_{j \geq 0} x^{10j}$ in this ring, (1) yields precisely the four stated residue classes. Substitute $2n$ and $2n - 1$ to obtain the two original assertions. For the latter, $2n - 1$ is the omitted odd residue class 5 modulo ten exactly when $n \equiv 3 \pmod 5$. □

The formal proof works with the integral series $x(\sum_{n \geq 0} x^n)^2$ and its actual third composition. It proves that reduction commutes with composition. The denominator cancellation in (1) uses a constructed unit inverse; it does not assume cancellation in an arbitrary ring. The final Lean statements retain the source's $n \geq 1$ and $k \geq 1$ restrictions where subtraction of natural numbers occurs.

3 The two cubic series: existence before congruences

3.1 A canonical integral family

Theorem 3.1. *For every $s \in \mathbb{Z}$, there is a unique normalized series $R_s \in \mathbb{Z}[[x]]$ satisfying*

$$R_s(3x^2) = 2x^2 + R_s(x)^2 + 2sR_s(x)^3. \tag{2}$$

Proof. For any normalized integral series U, define its residual by

$$E_s(U) = U^2 + 2sU^3 - U(3x^2) + 2x^2.$$

Every coefficient of $E_s(U)$ is even. Indeed, after reduction modulo two, Frobenius gives $U(x)^2 = U(x^2)$ and $3x^2 = x^2$, so the residual vanishes. Also $E_s(U) = O(x^3)$ for every normalized U.

Suppose $V - U = cx^r + O(x^{r+1})$, with $r \geq 2$. The quadratic difference has leading term $2cx^{r+1}$; the cubic difference has order at least $r + 2$; and $(V - U)(3x^2)$ has order at least $2r \geq r + 2$. Consequently

$$E_s(V) - E_s(U) = 2cx^{r+1} + O(x^{r+2}). \tag{3}$$

Starting with $U = x$, choose its degree-r coefficient to cancel the degree-$(r+1)$ coefficient of the residual. The needed division by two is exact by the preceding parity argument. Equation (3) shows that earlier cancellations remain valid. Each degree is thus determined by finitely many earlier coefficients, and the resulting integral series has zero residual in every degree.

For uniqueness, take the first degree $r \geq 2$ where two candidate solutions differ. Equation (3) would give a nonzero degree-$(r+1)$ difference of residuals, because $2c \neq 0$ in $\mathbb{Z}$. This contradicts both residuals being zero. □

The proof formalizes divisibility of the residual by two before using integer division. Thus the construction does not assume the existence or integrality of an OEIS sequence. The names used below denote the uniquely constructed series.

3.2 A389474 and a square-zero substitution

Let $R = R_1$ and let U be its compositional inverse. Such an inverse exists over $\mathbb{Z}$ for every normalized series: its degree-n coefficient is determined with leading multiplier one, recursively. Both inverse identities are proved in the reused formal-series infrastructure. Substituting $U(x)$ in (2) gives

$$R(3U(x)^2) = 2U(x)^2 + x^2 + 2x^3.$$

Applying U proves the original A389474 equation

$$U\big(x^2 + 2x^3 + 2U(x)^2\big) = 3U(x)^2. \tag{4}$$

Conversely, if a normalized U satisfies (4), substitute its inverse R and then apply R to recover (2) for $s = 1$. Therefore Theorem 3.1 also gives existence and uniqueness for the literal A389474 source equation.

Theorem 3.2. *The A389474 series satisfies*

$$U(x)^2 \equiv x^2 + 2x^3 \pmod 9.$$

In particular, $[x^n]U(x)^2$ is divisible by nine for every $n > 3$.

Proof. Work in $(\mathbb{Z}/9\mathbb{Z})[[x]]$ and put $S = U^2$, $Q = x^2 + 2x^3$. The source equation becomes $U(Q + 2S) = 3S$. Let V be the genuine normalized compositional inverse of U over this ring. Apply V to obtain

$$Q + 2S = V(3S).$$

The series $3S$ has zero constant coefficient and is square-zero: $(3S)^2 = 9S^2 = 0$. A normalized series has the form $V(t) = t + t^2H(t)$, so $V(3S) = 3S$. It follows that $Q + 2S = 3S$, hence $S = Q$, as required. No division by three is used. □

This establishes a full formal-series identity, including the low-degree terms, and hence strengthens the source's statement restricted to degrees above three.

3.3 Frobenius and A389475

For any $W \in \mathbb{Z}[[x]]$, prime p, and coefficientwise reduction $\overline{W} \in \mathbb{F}_p[[x]]$, the standard Frobenius identity is

$$\overline{W}(x)^p = \overline{W}(x^p). \tag{5}$$

One may prove it degree by degree: truncate W above the degree in question, apply the polynomial identity in characteristic p, and use $a^p = a$ in $\mathbb{F}_p$. The formal proof follows this finite-truncation argument. In particular, for every $k \geq 0$,

$$U(x)^{3k} \equiv U(x^3)^k \pmod 3.$$

This proves A389474's second conjecture, including the otherwise unneeded cases $k = 0$ and coefficient index zero. It is a universal algebraic fact, not a special property newly discovered for A389474.

Now let $V = R_{-1}$, which is the A389475 series. It satisfies

$$V(3x^2) = 2x^2 + V^2 - 2V^3. \tag{6}$$

Frobenius directly proves the first requested identity $V^3 \equiv V(x^3) \pmod 3$. Since $V(0) = 0$, reduction of (6) modulo three gives

$$0 = 2x^2 + V^2 - 2V^3,$$

or $V^2 = x^2 + 2V^3$ in $\mathbb{F}_3[[x]]$. Substitution of the Frobenius identity proves the second:

$$V(x)^2 \equiv x^2 + 2V(x^3) \pmod 3. \tag{7}$$

These are exactly the two assertions labeled (4.a) and (4.b) in the retrieved A389475 entry. The formal development also supplies the literal source-equation existence and uniqueness theorem.

4 A396102 and the inverse-iterate family

For each positive integer r, consider the normalized series equation

$$F_r^{\circ r}(x) = (1 + x)F_r^{\circ(r-1)}(x). \tag{8}$$

The case $r = 3$ is the original A396102 equation. We construct it through an auxiliary series, rather than assuming that a solution exists.

Lemma 4.1. *For every $r \geq 0$ and every commutative coefficient ring, there is a unique normalized series R_r satisfying*

$$R_r = x - R_r R_r^{\circ r}. \tag{9}$$

If $r \geq 1$, its compositional inverse F_r is the unique normalized solution of (8).

Proof. Define $T_r(G) = x - GG^{\circ r}$. It preserves normalization. If two normalized series agree through degree $n-1$, their iterates do also. The difference of the two products $GG^{\circ r}$ and $HH^{\circ r}$ has one factor divisible by x^n and another divisible by x in each term. Thus

$$G - H = O(x^n) \quad \Longrightarrow \quad T_r(G) - T_r(H) = O(x^{n+1}).$$

Starting at x and iterating T_r, every coefficient stabilizes after finitely many steps. The stabilized coefficients give a fixed point; the same order-increase implication proves uniqueness. This is a purely coefficientwise construction and works over arbitrary commutative rings.

Take the genuine compositional inverse F_r of R_r. Substitute $F_r^{\circ r}(x)$ into (9). Inverse cancellation gives

$$F_r^{\circ(r-1)} = F_r^{\circ r} - xF_r^{\circ(r-1)},$$

which is (8). Conversely, substitute $R^{\circ r}$ into (8), where $R = F^{-1}$. This gives $x = (1 + R^{\circ r})R$, hence (9). Uniqueness of R_r now implies uniqueness of F_r. □

Theorem 4.2. *For every $r \geq 1$,*

$$R_r(x) \equiv \frac{x}{1+x} \pmod{r}, \qquad F_r(x) \equiv \frac{x}{1-x} \pmod{r}.$$

Consequently, for $n \geq 1$ and $k \geq 0$,

$$[x^n]F_r^{\circ k}(x) \equiv k^{n-1} \pmod{r}. \tag{10}$$

Here $0^0 = 1$, consistently with the degree-one coefficient of the zeroth compositional iterate x.

Proof. Write $M_c(x) = x/(1-cx)$. Direct substitution gives $M_a \circ M_b = M_{a+b}$ over any commutative ring. Therefore $M_{-1}^{\circ r} = M_{-r} = x$ over $\mathbb{Z}/r\mathbb{Z}$. Moreover

$$x - M_{-1}(x)x = M_{-1}(x),$$

so M_{-1} satisfies the auxiliary fixed-point equation modulo r. The coefficientwise uniqueness in Lemma 4.1 identifies it with the reduction of R_r. Its actual inverse is M_1, identifying the reduction of F_r. Finally, $M_1^{\circ k} = M_k = x/(1-kx)$, whose degree-$n$ coefficient is k^{n-1} for $n \geq 1$. □

For $r = 3$ and $k = 1$, this proves the A396102 conjecture

$$[x^n]F_3(x) \equiv 1 \pmod{3} \qquad (n \geq 1).$$

The formal proof includes both directions of the correspondence with the original triple-composition equation, not merely a congruence for an auxiliary series. The formal parameterization uses $r + 1$ when stating the original equation so that its predecessor iterate is always a natural number. The coefficient theorem itself also has a well-defined $r = 0$ version in `ZMod 0`; no such extension is needed to settle the stated OEIS problem.

5 Known Stirling periodicity, now formalized

Let $S(n, k)$ be the Stirling numbers of the second kind, defined by $S(0, 0) = 1$, zero on the other boundary entries, and

$$S(n+1, k+1) = (k+1)S(n, k+1) + S(n, k).$$

For an arbitrary sequence of integers $c_0, c_1, \ldots$, put

$$b_n = \sum_{k=0}^{n} c_k k! S(n,k). \tag{11}$$

No positivity, boundedness, or recurrence condition is imposed on the weights. The usual formal exponential identity in $\mathbb{Q}[[x]]$ identifies (11) as the coefficient sequence of

$$\sum_{n\geq 0} b_n \frac{x^n}{n!} = G(e^x - 1), \qquad G(t) = \sum_{k\geq 0} c_k t^k \in \mathbb{Z}[[t]].$$

Theorem 5.1. *For every $m > 0$ and $n \geq m$,*

$$b_{n+\varphi(m)} \equiv b_n \pmod{m}.$$

The least positive period of the tail beginning at index m exists and divides $\varphi(m)$.

Proof. We first show that every natural base a satisfies

$$a^{n+\varphi(m)} \equiv a^n \pmod{m} \qquad (n \geq m). \tag{12}$$

It suffices to check each prime power p^e dividing m. If p divides a, then $e \leq p^e \leq m \leq n$, so both powers vanish modulo p^e. If p does not divide a, Euler's theorem applies modulo p^e, and $\varphi(p^e)$ divides $\varphi(m)$. This proves (12) by prime-power divisibility.

The power expansion is

$$a^n = \sum_{k=0}^{a} \binom{a}{k} k! S(n,k) \qquad (a \leq n). \tag{13}$$

It follows from the Stirling recurrence and the usual recurrence for falling factorials. Fix $n \geq m$. Strong induction on $k < m$ shows that $k!S(n+\varphi(m),k)$ and $k!S(n,k)$ are congruent modulo m: apply (13) at $a = k$ to the two exponents, cancel all terms with index below k using the induction hypothesis, and note that $\binom{k}{k} = 1$. This argument makes no division by $k!$ modulo m.

For $k \geq m$, the factor $k!$ is divisible by m. Thus (11) reduces to the fixed finite sum with $k < m$. The preceding columnwise congruence proves the claimed period for arbitrary integral weights. Since $\varphi(m) > 0$, the tail has at least one positive period.

For completeness, let d be its least positive period and write any period $p = qd + r$, $0 \leq r < d$. Repeated use of the d-periodicity shows that r is also a period. Minimality forces $r = 0$, so d divides p. Apply this with $p = \varphi(m)$. In Lean this last step uses the standard minimal-period theorem for the shift map on the entire tail. □

Prior work and exact formal scope. Danesh's 26 August 2026 preprint proves the general integral-substitution result with the stronger eventual period $\lambda(m)$, the Carmichael function [5]. It therefore already covers Theorem 5.1 as an eventual-period assertion and all seven applications below. Our displayed bound $n \geq m$ is explicit; no priority claim is made for this elementary threshold. Schreyer earlier treated Fubini and r-Fubini numbers with a Carmichael bound [6].

The Lean theorem is the fully quantified finite-sum statement (11), with exact definitions for every named weight. The identity involving the formal exponential series is explained mathematically here and is *not separately claimed as a Lean theorem.* For the named applications, the OEIS entries provide the same Stirling-transform formulas or equivalent partition generating functions. Numerical prefix checks further corroborate their indexing.

Entry	Weight c_k in (11)
A052895	Catalan number $\frac{1}{k+1}\binom{2k}{k}$
A064618	$k!$
A354242	$\binom{2k}{k}$
A167137	Number of integer partitions of k
A305550	Number of partitions of k into distinct parts
A306082	Number of partitions of k into square parts
A080253	$1, 3, 7, 17, \ldots$, with $c_{k+2} = 2c_{k+1} + c_k$

Table 2: Seven credited applications of an already proved general theorem.

The partition weights in Lean are actual finite cardinalities of `Nat.Partition`, with distinctness or square-part restrictions where appropriate. The Pell weight is an explicit recurrence. No named weight is left as an unspecified function assumed to match an OEIS sequence. We do not address the separate continued-fraction conjecture in A354242 or the separate power/logarithm integrality conjecture in A167137.

6 Formal verification and reproducibility

The combined package pins Lean 4.19.0 and Mathlib revision

`c44e0c8ee63ca166450922a373c7409c5d26b00b.`

The additional results reside in `lean/More/`. The principal module groups are:

- `RationalParity.lean`: the actual third composition, characteristic-two denominator identity, and exact parity conclusions.
- `CubicFrobenius.lean`, `CubicReversion.lean`, and `CubicConsequences.lean`: Frobenius, integral construction, source-equation equivalence, and all four cubic congruences.
- `InverseIterate.lean`: the auxiliary fixed point, genuine inverse, original source equation, uniqueness, and congruences.
- The Stirling, power-periodicity, and tail-periodicity modules: the finite transform theorem, actual named weights, and least periods.

The release checker has a fixed list of **130 theorem endpoints**: the original 67 and 63 added endpoints, including construction and transport results as well as named consequences. This is a proof-audit count, not a count of open problems. The checker removes the first-party build cache, rebuilds the project, compiles explicit semantic contracts, prints fully elaborated theorem types, and checks every endpoint's transitive axiom dependencies. The only permitted axioms are `propext`, `Classical.choice`, and `Quot.sound`. No `sorry`, `admit`, `native_decide`, or custom mathematical axiom is used in the proof sources.

`More/Contracts.lean` states the source equations, definitions, coefficient domains, and unbounded conclusions independently of the exported theorem types. The old contracts are retained unchanged within the expanded contract file. Mechanical success establishes the stated Lean propositions relative to Lean's foundations; checking the translation from OEIS prose to those propositions is a separate semantic responsibility. AI reviewers examined those correspondences, and the package records their findings. Independent human review remains pending.

Finite calculations. Independent exact-integer scripts construct coefficients from the original equations, rather than from the desired modular patterns. The cubic check runs through degree 40 and tests both equations through degree 41. The inverse-iterate check covers parameters 1 through 8 and iterates 0 through 8 through degree 40. The parity check uses direct composition, a separate coefficient formula, and the rational recurrence, including parity through degree 2000. The Stirling check covers moduli 1 through 128, signed as well as positive weights, and 81,376 period comparisons. JSON files record their exact scopes. These calculations are corroboration; the Lean proofs establish the claims for every allowed index.

Reproduction. With the pinned Lean toolchain and Python 3.11 or later installed, run the following from the extracted directory:

```
cd lean
lake exe cache get
lake build
cd ..
python3 scripts/check_formalization.py
```

The checker rejects missing endpoints, changed dependency pins, warning/error output, extra axioms, and first-party source changes during validation. It also verifies the previous archive's fixed SHA-256 and the unchanged previous Lean proof modules. The generated report and logs in `reproducibility/lean_complete/` are the authoritative record of the delivered run.

7 What the combined archive retains

The previous package's compositional-congruence manuscript and all of its Lean modules are included. Those cover all four A396099 claims; the A396798 iterate assertions and the correction of a false starting phase; A396807 including integer iterates; the A393856–A393859 scaled-inverse assertions; and the original A396843-family acceptance theorems. Their complete proofs remain in the previous manuscript and source. This new document adds to that work rather than revising its mathematical conclusions.

For exact preservation, the archive also contains the original ZIP at

`prior_work/oeis_congruence_proofs_Dev_Goyal_v1.zip`.

Its SHA-256 is

`04f347b0e116a1c92cd7a1f21f38a37ae3d`
`fb71c733dd5f55769accd0714489c`

The line break is typographical; the checksum is one 64-character string in the validation manifest. The package preserves the author details from the supplied repository and does not reuse the earlier work's DOI as a DOI for this expanded release.

The general Stirling recurrence and part of its power-expansion proof were adapted from a later Apache-2.0 Mathlib module. Its copyright, original authors, exact revision, license, and change notice are retained under `third_party/mathlib/`. This upstream reuse is separate from the AI-generated modular transfer and sequence-specific proofs.

References

[1] P. D. Hanna, OEIS A396093, `https://oeis.org/A396093`. Accessed 24 October 2026.

[2] P. D. Hanna, OEIS A389474, `https://oeis.org/A389474`. Accessed 24 October 2026.

[3] P. D. Hanna, OEIS A389475, `https://oeis.org/A389475`. Accessed 24 October 2026.

[4] P. D. Hanna, OEIS A396102, `https://oeis.org/A396102`. Accessed 24 October 2026.

[5] P. Danesh, *Eventual periodicity of weighted Stirling transforms and two conjectures of Bala*, version 1, 26 August 2026, Theorem 4.1 and Corollary 4.2. `https://doi.org/10.33774/coe-2026-4d6jr`. Preprint; not peer-reviewed by Cambridge University Press at posting.

[6] B. Schreyer, *Rigged Horse Numbers and their Modular Periodicity*, arXiv:2409.03799v2, 3 December 2024. `https://arxiv.org/abs/2409.03799`.

[7] OEIS entries A052895, A064618, A354242, A167137, A305550, A306082, and A080253. Each is available at `https://oeis.org/` followed by its identifier. Accessed 24 October 2026; precise formulas and status are in the source ledger.

[8] D. Goyal, *oeis-a396843-proof*, supplied source repository, revision `995e55adcbe06b2a913e24f12ff73a7faf362f06`. `https://github.com/dvgyl/oeis-a396843-proof`.

[9] Mathlib contributors, formal proof library for Lean. Pinned dependency and adapted upstream module revisions are recorded in `lean/lake-manifest.json` and `third_party/mathlib/NOTICE.md`, respectively.